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13 March 1985
SUPERSEDING
MIL-D-52923A(ME)
28 October 1981

MILITARY SPECIFICATION

DISTRIBUTOR, WATER, SEMITRAILER MOUNTED, 6000 GALLON

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a semitrailer mounted, 6000 gallon water distributor.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

ZZ-H-451	- Hose, Fire, Woven-Jacketed, Rubber- or Cambric-Lined, with Couplings.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3825

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-771	- Distributor, Bitumen and Water, Packaging of.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-V-16721	- Valves, Foot, Low Pressure (for Water).
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-T-46786	- Tanks, Fuel, Engine: General Requirements for.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-209	- Slinging and Tiedown Provision for Lifting and Tying Down Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Material.
MS27029	- Coupling Half, Quick Disconnect, Cam & Locking Type, Plug Dust, Type X.
MS75021	- Connector, Receptacle, Electrical- 12 Contact, Intervehicular, 28-Volt, Waterproof.

(Copies of specifications and standards required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein.

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR

Occupational Safety Health Act.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402)

DEPARTMENT OF TRANSPORTATION

Federal Motor Vehicle Safety Standards.
Federal Motor-Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Sections 5 and 6, Welding Qualifications.
D14.3 - Earthmoving and Construction Equipment Welding, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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HYDRAULIC INSTITUTE

Standards For Centrifugal, Rotary and Reciprocating Pumps.

(Applications for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

CALIFORNIA AIR RESOURCES BOARD REGULATIONS

(Application for copies should be addressed to the California Air Resources Board, 1709 11th Street, Sacramento, CA 95814.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

STEEL STRUCTURE PAINTING COUNCIL

SSPC-SP6 - Commercial Blast Cleaning.

(Application for copies should be addressed to the Steel Structure Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a semitrailer mounted, 6000 gallon capacity water distributor. The distributor shall be equipped with a diesel engine driven pump, remotely controlled by a single operator from the truck-tractor cab. The semitrailer mounted distributor shall be certified as complying with the Department of Labor (OSHA) Regulations and applicable DOT Standards and Regulations in effect on the date of manufacture.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.3 Production distributors. All distributors furnished under this specification shall be physically and mechanically identical to the first article distributor approved by the Government.

3.4 Construction.

3.4.1 General. The distributor shall consist basically of a tank and spray system mounted on a semitrailer chassis as specified herein.

3.4.2 Performance. The distributor shall:

a. Be compatible with the Military M123A1C, M916, and M920 truck tractors. When coupled to any of the three truck tractors, the vehicle shall be capable of making interference free full left and right turns while cresting hills which generate pitch angles of up to plus or minus twelve degrees (±12°) between truck tractor and distributor.
b. Travel, with maximum payload and without payload on improved roads at speeds up to 50 miles per hour, on unpaved roads at speeds up to 30 miles per hour, and on loose fill at speeds up to 5 miles per hour without evidence of malfunction, permanent deformation or damage. Loose fill is defined as newly placed uncompacted fill or in place material which has been ripped, tilled or scarified to a depth of at least six inches.
c. Comply with Department of Transportation Motor Vehicle Safety Regulations for towed vehicle tracking tolerances.
d. Be capable of assuming a 90 degree angle to the towing vehicle without interference or cramping, sideslipping, or damage to the semitrailer or the towing vehicle.
e. When coupled or uncoupled to the truck tractor, be capable of filling the tank from an outside source using the pump and engine.
f. When coupled to the truck tractor, be capable of pumping from the tank through the nozzles.
g. When coupled or uncoupled to the truck tractor, be capable of pumping from the tank through a 1-1/2-inch outlet.
h. Be capable of draining the tank without water going through the pump.

3.4.3 Weights and dimensions. The distributor shall conform to the following:

Weight on fifth wheel (maximum, loaded)	30,000 pounds
Overall length (maximum)	38 feet
Overall height	10 feet
Overall width	96 inches
Upper fifth wheel height (as measured with the plate in a horizontal plane, tank empty with the trailer on a level plane)	62 inches (±1 inch) (see 3.8.1)

Interchange coupling dimensions shall be compatible with the M123A1C, M920, and M916 truck tractors.

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3.4.4 Environmental conditions. The distributor shall be capable of functioning as a water distributor in temperatures from +40° F to 120° F.

3.4.5 Stability. The distributor shall be capable of negotiating in both directions at least a 10 percent side slope when empty and full.

3.4.6 Fording. The distributor shall start and operate during and after fording operation in fresh water to a depth of 30 inches without damage and without leakage of water into any reservoir including reservoirs containing lubricants.

3.4.7 Maintenance.

3.4.7.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.2 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using general purpose tools normally associated with equipment of this nature. After completion of any maintenance operation, the distributor shall operate as specified herein.

3.4.7.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.5 Material deterioration and control. The distributor shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.5.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.5.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components or sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.5.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the distributor may be newly fabricated from recovered materials to the maximum extent practicable, provided the distributor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the distributor.

3.6 Tank. The tank shell, baffles, heads, manhole, vents, and outlets shall be fabricated from steel or type 304 stainless steel. If steel is used, the interior of the tank surface shall be free of weld spatter, oil and grease prior to sandblasting. The interior metal surfaces shall be sandblasted in accordance with SSPC-SP-6 with a surface profile not to exceed 3 mils. The interior surfaces shall be treated with a self priming non-toxic coal tar material suitable for water tanks within 8 hours of sandblasting. Coating thickness shall be not less than 20 mils dry film thickness average with a minimum thickness of 16 mils. All areas less than 16 mils shall be recoated to achieve 20 mils. Shop prime metal surfaces 2 mils dry film thickness if top coating cannot be applied within the 8 hour period. The manufacturer shall provide a steel sheet thickness for the tank cylinder which is structurally adequate for his design; however, thickness shall in no case be less then 0.1644 inch (8 gage). Sheet steel used in the heads shall not be less than 0.1644 inch (8 gage). The heads shall be concave, convex, or reinforced flat plates and all seams shall be welded and watertight. Each tank shall have transverse surge plates made of not less than 0.1345 inch (10 gage) sheet steel. The spacing between surge plates and between surge plates and tank ends shall be reinforced to withstand the surge pressure; and shall permit free flow of water at the bottom, venting at the top, and access to all sections of the tank. Holes in the surge plates shall be staggered. The tank shall be provided with a manhole not less than 20 inches in diameter. The manhole shall be located at the top of the tank and shall be equipped with a hinged, watertight, quick-clamp-type cover. The low water level indicator mechanism shall be protected from personnel entering and exiting the tank through the manhole. A ladder or rungs shall be provided on the tank for access from the ground to the manhole. All operator access areas shall be equipped with non-skid decking surfaces, and be equipped with handholds that permit use of arctic mittens. A 3-inch unrestricted opening for overflow and for venting to the atmosphere at all times shall be provided and shall be protected against surge discharge. The 3-inch opening shall be located not more than 2 feet from the transverse centerline of the tank. The tank capacity shall be not less than 6,000 gallons. The finished tank shall contain no materials nor substances which might leak out or disintegrate and cause the water to become toxic. The material shall have no adverse affect on the health of personnel when used for its intended purpose.

3.7 Spray system. The spray system shall consist of, but not be limited to, a self-priming centrifugal pump driven by a liquid cooled diesel engine and necessary piping and controls to permit the distributor to function as specified herein.

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3.7.1 Pump. A centrifugal pump which is self-priming after initial filling shall be provided. When operating against a total head of 90 feet and with a 10 foot static suction lift, and when operating with the suction and discharge ports provided, the pump shall deliver water at a rate of not less than 600 gallons per minute. When operating against a total head of 125 feet and 10 foot static suction lift and when operating with the suction and discharge ports provided the pump shall deliver water at a rate of not less than 200 gallons per minute. The horsepower demand of the pump under the above conditions shall not exceed the intermittent horsepower rating of the engine. The pump assembly shall operate for not less than 100 hours without a failure when operated as follows:

a. At a capacity commensurate with the continuous horsepower rating of the engine.
b. At maximum pump capacity if the continuous horsepower rating of the engine exceeds the maximum power required by the pump.

3.7.2 Diesel engine. For the purpose of selecting the engine, the water distributor shall be classified as a class III end item in accordance with MIL-STD-1410. The engine, including all systems, components and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. The engine shall be liquid cooled. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the Environmental Protection Agency and the California Air Resources Board regulations governing control of air pollution for new motor vehicles and new motor vehicle engines, in effect for this type of equipment on date of manufacture. The engine shall meet all operational requirements and conditions specified in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits. The engine shall be furnished as a complete power unit, including housing, instruments, instrument panel, controls, battery and battery box. Exhaust gas or discharges from the engine shall be directed so that they do not endanger personnel. All hot surfaces of the engine, including exhaust pipes, exposed to contact by personnel or which create a fire hazard, shall be guarded or insulated.

3.7.2.1 Engine air induction system. The air induction system shall include a heavy-duty, dry type air cleaner and an air cleaner restriction indicator in accordance with MIL-STD-1410.

3.7.2.2 Cranking and charging system. The engine shall be furnished with a 12-volt electric cranking and charging system in accordance with MIL-STD-1410. The battery charging alternator shall be self-regulated, heavy-duty and shall have a capacity of not less than 25 amperes. The battery shall be size 6-TM. An ammeter shall be mounted on the engine instrument panel.

3.7.2.3 Engine safety controls. Engine control systems, of the engine shut-off type, shall be furnished in accordance with MIL-STD-1410 as follows:

a. High coolant temperature.
b. Low lubricating oil pressure.

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3.7.2.4 Manual throttle control. A manual throttle control shall be furnished in accordance with MIL-STD-1410 and shall be mounted on the instrument panel.

3.7.2.5 Fuel tank. A fuel tank with a capacity of not less than 12 gallons shall be provided. The fuel tank shall conform to MIL-T-46786, type III, except that the tank shall be exempt from the fire test. The fuel tank shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Adequate shielding may be provided to attain the same purpose. The lettering "Diesel Fuel" shall be stenciled in one inch letters in a conspicuous location near the fuel tank opening.

3.7.2.6 Muffler. The engine shall be equipped with a muffler to reduce the noise level emitted by the engine. Vertical exhaust pipe shall be equipped with a self-opening and closing rain cap.

3.7.3 Piping. Two adjustable nonferrous or noncorrosive spray nozzle shall be provided and shall be mounted on the rear of the distributor. Each nozzle shall be independently controlled (on-off) from the remote control box, and from the rear platform. Each nozzle shall be manually adjustable. Nozzle adjustment shall permit an overlapping fan-like spray pattern covering a swath of at least 70 feet (not less than 35 feet to either side of the distributor longitudinal centerline). Single nozzle operation shall permit covering a swath up to 30 feet (not less than 15 feet to either side of a line through the center of each nozzle parallel to the longitudinal centerline of the distributor). Each nozzle shall be mechanically/spring closed and independently opened with compressed air. The piping shall be sized and designed to permit pumping at a rate of not less than 600 gallons per minute. An operating instruction plate shall be provided describing step-by-step procedures required to perform all operations specified herein and shall include a flow and valve diagram. The plate shall be permanently affixed to the distributor. The plate shall conform to MIL-P-514, type III, composition C, photosensitive anodized aluminum. The materials and treatment used on surfaces in contact with water shall be nontoxic to the health of personnel when the distributor is used to transport potable water.

3.7.4 Spray accessories. The distributor shall be provided with the following:

a. Two 15-foot sections of suction hose 4-inches in diameter. Provisions shall be made to connect the sections to each other by use of quick-disconnect couplings. A foot valve in accordance with MIL-V-16721 shall be provided for use with the suction hose.
b. Permanently mounted storage rack(s) shall be provided on the distributor to accommodate the two 15-foot sections of suction hose. The storage rack(s) shall prevent dirt and mud thrown by the tires of the towing unit from entering the suction hoses, or dust plugs shall be provided. When dust plugs are provided they shall be in accordance with MS27029-18 and they shall be chain secured to the suction hoses.

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c. A 4-inch quick-disconnect coupling provided on the suction side of the pump to which the suction hose can be connected. A chain secured dust cap shall be provided. The couplings shall be accessible to an operator standing on the ground.
d. A fire hose in accordance with ZZ-H-451. The hose shall be not less than 50 feet in length. Fittings shall be provided to connect the hose to the 1-1/2 inch outlet (see 3.4.2g).
e. A fire hose nozzle (with on/off control) compatible with the hose fittings shall be provided.
f. A removable, galvanized, inlet screen to protect the pump from particles greater than 1/8 inch during tank loading.
g. All necessary valving, fittings, and dust caps shall be provided. Means shall be provided to store the accessories on the distributor. Each accessory shall remain in its place during any operation specified herein.
h. Instructions for water distribution rates shall be provided with each distributor. The instructions shall be encased in waterproof laminated plastic. The instructions shall provide the operator with the water distribution rate in gallons per 100 foot length pattern, correlated to the following variables:

    (a) First and second tractor transmission geared speeds at 100 percent and 50 percent governed engine rpm.
    (b) Water pump speeds of 100 percent and 50 percent of engine governed rpm.
    (c) Per nozzle at maximum opening size, using 1 and 2 nozzles. The distribution rate chart shall be within 15 percent of actual or measured rate for each variable.

3.7.5 Remote control. The distributor controls shall be incorporated in a common panel. The panel shall be connected to the semitrailer with a jacketed cable adequately supported and of sufficient length to permit temporary placement of the panel in the cab of the towing vehicle and permit normal distributor operations, including right and left jackknifes up to 90 degrees. The remote control cable shall be located in a protected location and shall be supported with cushioned clamps through the semitrailer length at intervals of not more than 3 feet. All jacketed cable components such as pneumatic lines, fittings, electric cable, connectors, etc., shall be suitable for use over a temperature range of -40° to +200° F. Pneumatic lines shall be in accordance with SAE J1402. Pneumatic lines shall be provided with spring guards or similar devices to reduce tendency of lines to crimp or collapse at line fittings. The controls shall permit independent on-off operation of each spray nozzle valve. Control of the nozzle valves and engine throttle control shall be accomplished by air pressure derived from the truck-tractor air supply. The panel shall contain an engine tachometer, a distributor tank low water level indicator light, and controls to allow pump engine start up, shutdown and increase and decrease in engine rpm's. The low water level indicator light shall indicate when the water level in the tank is 500 gallons or lower. Each control shall be clearly and permanently identified. Self-adhering tape for identification is not permitted. Means shall be provided to store the remote control on the distributor when not in use. A bracket shall be provided to firmly mount the panel in the

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truck-tractor. The bracket shall permit quick installation and removal of the panel. All necessary hardware and installation instructions for the panel and bracket shall be provided with each distributor.

3.7.6 Engine ignition switch. The engine shall be capable of being independently started and stopped from either the remote control panel or the engine control panel.

3.7.7 Engine ignition switch pilot light. The remote control panel and the engine control panel shall have a red pilot light to indicate when the respective ignition switch is in the on position.

3.8 Semitrailer chassis. The semitrailer shall have rigidity and strength to withstand operations as specified herein without permanent deformation or damage and shall conform to the requirements herein.

3.8.1 Upper fifth wheel. The semitrailer shall be furnished with an upper fifth wheel and kingpin which provides for the fifth wheel height specified in 3.4.3. The upper fifth wheel plate shall be of the full width type and shall cover a 36-inch diameter lower fifth wheel. The upper fifth wheel assembly shall conform to DOT MCSR 393.70. The plate shall include a front coupling ramp with a minimum of 2-inch vertical height. Design shall prevent collection of water in the upper fifth wheel. The upper fifth wheel shall be equipped with a kingpin in accordance with SAE J848. External SAE lubrication fittings shall be provided to permit lubrication of road and curb sides of the tractor-semitrailer combination while coupled.

3.8.2 Semitrailer front supports. The front supports (landing gear) shall be the vertical and retractable two-speed type with legs operated and controlled by a handcrank. Each leg shall be braced and shall be capable of supporting its portion of the fully loaded semitrailer. The supports shall withstand normal usage without permanent deformation. Legs shall be provided with metal pads to reduce contact pressure. Supports for the crank extension shaft and clips for holding the crank when folded shall be provided. Not less than 12 inches of upward and 7 inches of downward travel from the height of the level trailer shall be provided.

3.8.3 Running gear. A tandem axle and suspension system shall be provided. The rated capacity of the axles and suspension shall be at least equal to the load imposed by the fully loaded distributor.

3.8.3.1 Axles. The axles shall be oil lubricated and provided with visual oil level windows.

3.8.3.2 Tires and wheels. All tire and wheel assemblies shall be in conformance to the Tire and Rim Association Yearbook. The wheels shall be of the demountable disc type. All wheels and tires furnished shall be identical. Dual wheels shall be provided for each axle. The recommended tire pressure shall be stenciled near each tire in 1 inch high letters.

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3.8.3.3 Suspension. The suspension shall be of the flexible leaf spring type.

3.8.4 Rear end protection and wheel splash protection. The semitrailer shall be provided with the following:

a. Rear end protection in accordance with Department of Transportation Motor Carrier Safety Regulations.
b. Rear fenders.
c. Splash guards in accordance with SAE J682.

3.8.5 Air brake system. The distributor shall be equipped with a service and emergency parking air brake system to include a spring parking brake system. The brake system shall conform to all applicable requirements of DOT FMVSS and DOT MCSR except that FMVSS 121 shall not apply.

3.8.6 Gladhands. Gladhands shall be provided in accordance with SAE J318, color coded yellow for service and red for emergency. Spring loaded dust covers for both gladhands shall be provided. The location of the gladhands shall be in accordance with SAE J702.

3.8.7 Electrical. Electrical wiring system shall be 12 volt potential and as specified herein. Electrical system shall be in accordance with Motor Carrier Safety Regulations. The electrical system shall be designed to withstand the vibration and shock induced by highway, unpaved road, and loose fill travel of the distributor without loss of continuity or damage.

3.8.8 Lighting system. The complete lighting system shall conform to Motor Carrier Safety Regulations. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on rub rails or vehicle bumper. All wiring shall be in conduit with threaded joints having waterproof connections. Conduit shall be provided with vents located in weather protected locations. Turn signal lamps shall be in accordance with SAE J588. All wiring shall be properly routed and anchored to protect against abrasion, snagging and tearing away. Wiring shall be in accordance with SAE J555.

3.8.9 24 volt receptacle system. In addition to the 7-contact SAE J560 receptable, the front of the semitrailer shall be equipped with one 12-contact receptacle conforming to MS75021-1 with spring-loaded cover. The 12-contact receptable shall be provided with resistance to each circuit to reduce the voltage of a tactical military design towing vehicle from 24-volts to 12-volts. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12-volts directly to the semitrailer circuits. The 24-volt 12-contact receptacle shall be connected through the necessary resistors into the following 12-volt trailer circuits as follows: (Letters depict the 24-volt contact terminals and the numbers refer to the SAE J560 receptacle).

a. Do not connect.
b. Connect to circuit No. 3.

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c. Do not connect.
d. Connect to circuit No. 1.
e. Connect to circuits No. 2 and No. 6.
f. Do not connect.
h. Do not connect.
j. Connect to circuit No. 5.
k. Connect to circuit No. 7.
l. Do not connect.
m. Do not connect.
n. Do not connect.

Circuits b and j on tactical truck tractors are combination stop and turn indicator circuits. The normal 12-volt signal lights will function both as turn signals and stop lights and the normal 12-volt stop lights will not be in operation when the trailer is connected to a towing vehicle with a 24-volt power supply. Because of this condition the No. 4 on stop light circuit is not connected to the 24-volt 12-contact receptacle.

3.9 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.9.1 Lubricants. The distributor shall operate as specified herein when lubricated with lubricants conforming to MIL-G-10924, MIL-L-2104, and MIL-L-2105.

3.9.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.9.3 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.9.4 Lubrication. The distributor shall be assembled, adjusted, run-in, and delivered with lubricants specified in 3.9.1. Bearings shall be cleaned before lubrication because military grease is not necessarily compatible with other grease. The distributor shall be tagged in a conspicuous place to indicate that military grease has been used.

3.10 Slinging attachments. Permanently affixed attachments that enable the distributor to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209, for type II equipment, class 1 or 3 provisions. Slinging attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.11.

3.11 Equipment tiedown attachments. Permanently affixed attachments that permit fastening the distributor to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209, for type II equipment, class 2 or 3 provisions.

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3.12 Treatment and painting. The portions of the distributor normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Final color shall be forest green.

3.13 Toolbox. A lockable weather protected storage compartment(s) or toolbox of sufficient size to hold all tools and accessories shall be provided.

3.14 Identification plates and marking. Each distributor shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, vehicle identification number, weight, date of manufacture and date of delivery. The identification plate shall be permanently affixed to the distributor in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contacting officer. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly. The distributor and each attachment shall be identified in accordance with the contractor's standard method.

3.15 Transportation data plate. The transportation data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, and, in addition, shall show the silhouette of the distributor in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.16 Stenciling. The gross weight of the distributor shall be stenciled on each side so as to be discernible. The words "LIFT" and "TIEDOWN", as applicable, shall be stenciled adjacent to their respective attachments using block letters not less than 1 inch high. Stenciling shall conform to MIL-T-704.

3.17 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

3.18 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.19 Government-loaned property. A military M920, 20 ton, 8x6 truck-tractor will be loaned by the Government (see 6.4).

3.20 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts and castings shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free from all burrs; sharp edges; deleterious die, tool and metal handling marks; rust, scale, and cracks. All materials used shall be free from defects. Improper manufacturing or assembling practices, defective components or parts and assemblies, which have been modified to overcome deficiencies shall not be furnished.

3.20.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter.

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Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the distributor to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible, and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. The welding process used in fabrication of the distributor shall be at the option of the contractor.

3.20.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by one of the following listed codes or the manufacturer's commercial code, for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Welding Qualifications, Sections 5 and 6.
ASME - Welding Qualifications of Section IX.
AWS/ANSI D14.3 - Earthmoving and Construction Equipment Welding, Qualification, Section 5.

Certification shall be made available for review by the contracting officer or designated representative. Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

3.20.1.2 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the distributor, the distributor shall conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.20.1.3 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations
b. Tack welds shall be subject to the same quality requirements as final welds.

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c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and distributor.

3.21 Noise level. The noise level shall not exceed MIL-STD-1474, 5.1.1.1, category D [85 dB(A)], at the operators' position or other occasionally occupied positions, when tested in accordance with 4.5.2.15. If technical infeasibility is established via MIL-STD-1474, 5.1.1.2, or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85 db(A) limit. The precautions of MIL-STD-1474 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign. Technical manuals shall contain these warnings.

3.22 Certification. Unless otherwise specified (see 6.2), the contractor shall provide certification that the distributor complies with the requirements of OSHA and DOT regulations (see 3.1), DOT tracking tolerance in 3.4.2c, environmental condition in 3.4.4, tank material/coating in 3.6, fuel tank in 3.7.2.5, non toxic surfaces in 3.6 and 3.7.3, and component temperature range in 3.7.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all sections 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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c. Inspection comparison (see 4.5).
d. Toxicological data and formulations inspection (see 4.6).
e. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. Each distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Tests marked by "X" in column I of table I shall be performed. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection. Each vehicle, without payload and with lighting system on, shall be operating for a distance of not less than fifteen miles. After the fifteen mile run, the vehicle shall be examined for the defects specified in 4.5.1. Each vehicle, completely assembled, adjusted, and serviced, shall then be subjected to the tests marked by an X in column 2 of table I.

TABLE I. Test schedule.

First article	Quality conformance	Test	Requirement Paragraph	Test paragraph
1	2	3	4	5
X	X	Production run-in.1/	3.9.4	4.5.2.1
X		Travel.	3.4.2	4.5.2.2
X	X	Axle and suspension check.	3.8.3, 3.8.3.1, 3.8.3.3	4.5.2.3
X	X	Air brake system.	3.8.5, 3.8.6	4.5.2.4
X	X	Electrical system check.	3.8.7 thru 3.8.9	4.5.2.5
X		Fording check.	3.4.6	4.5.2.6
X		Stability check.	3.4.5	4.5.2.7
X		Slinging provisions.	3.10	4.5.2.8
X		Tiedown provisions.	3.11	4.5.2.9
X		Vehicle weight.	3.4.3	4.5.2.10
X	X	Pumping system operation.	3.4.2	4.5.2.11
X		Pump performance test.	3.7.1	4.5.2.12
X		Pump operation test.	3.7.1	4.5.2.13
X		Tracking tolerance, jack-knife, and landing leg test.	3.4.2	4.5.2.14
X		Sound level measurement.	3.21	4.5.2.15
X		Spray distribution rate test.	3.7.4	4.5.2.16
X		Spraying mode test	3.7.5	4.5.2.17
X		Compatibility check.	3.4.2	4.5.2.18

1/ The production run-in test shall be conducted before any other test.

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4.4.1 Examination. Each distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Individual. Each distributor shall be subjected to the tests marked by an X in column 2 of table I. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Each distributor shall be examined as specified herein for the following defects:

Major	Defect	Requirement paragraph
101.	Materials, dimensions, and weights not as specified.	3.4.3
102.	Maintainability and interchangeability not as specified.	3.4.7
103.	Materials not resistant to corrosion and deterioration, for the applicable storage and operating environments.	3.5
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.5.1
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.5.2
106.	Tank not as specified.	3.6
107.	Tank interior coating thickness not as specified on all interior surfaces.	3.6
108.	Spray system not as specified.	3.7
109.	Diesel engine not as specified.	3.7.2
110.	Engine air induction system not as specified.	3.7.2.1
111.	Cranking and charging system not as specified.	3.7.2.2
112.	Engine safety controls not as specified.	3.7.2.3
113.	Manual throttle control not as specified.	3.7.2.4
114.	Fuel tank not as specified.	3.7.2.5
115.	Muffler not as specified.	3.7.2.6
116.	Piping not as specified.	3.7.3
117.	Spray accessories not as specified.	3.7.4
118.	Remote control not as specified.	3.7.5
119.	Engine ignition switch not as specified.	3.7.6
120.	Engine ignition switch pilot light not as specified.	3.7.7
121.	Semitrailer chassis not as specified.	3.8
122.	Upper fifth wheel not as specified.	3.8.1
123.	Semitrailer front supports not as specified.	3.8.2
124.	Running gear not as specified.	3.8.3
125.	Axles not as specified.	3.8.3.1
126.	Tires and wheels not as specified.	3.8.3.2
127.	Suspension not as specified.	3.8.4
128.	Rear end protection and wheel splash protection not as specified.	3.8.4

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129.	Air brake system not as specified.	3.8.5
130.	Gladhands not as specified.	3.8.6
131.	Electrical system not as specified.	3.8.7, 3.8.8, 3.8.9
132.	Lubrication not as specified.	3.9 thru 3.9.4
133.	Slinging and tiedown provisions not as specified.	3.10, 3.11
134.	Painting, marking and data plate not as specified.	3.2, 3.8, 3.12, 3.14, 3.15 and 3.16
135.	Toolbox not as specified.	3.13
136.	Technical publications, repair parts, and maintenance tools not as specified.	3.17, 3.18
137.	Workmanship not as specified.	3.20
138.	Welding not as specified.	3.19.1
139.	Certification not as specified.	3.6, 3.22

4.5.2 Test.

4.5.2.1 Production run-in. Start and operate each distributor for a period not less than one hour to assure operation of all components and to allow for minor adjustments. All controls shall be manipulated. Evidence of leakage, malfunction, permanent deformation or inability of the distributor to operate as specified shall constitute failure of this test.

4.5.2.2 Travel. The distributor, with lights on, shall be operated as specified in table II. The distributor shall be completely assembled and serviced. Test course or area shall be selected by the contractor. Evidence of malfunction, permanent deformation or damage shall constitute failure of this test.

TABLE II. Mileage and speed tests.

Type of course	Mileage and Speed	Payload
Primary highway	100 miles at variable speeds up to 50 mph	Max. payload
Primary highway	100 miles at variable speeds up to 50 mph	W/o payload
Unpaved road	50 miles at variable speeds up to 30 mph	Max. payload
Unpaved road	50 miles at variable speeds up to 30 mph	W/o payload
Loose fill	25 miles at variable speeds up to 5 mph	Max. payload
Loose fill	25 miles at variable speeds up to 5 mph	W/o payload

4.5.2.3 Axle and suspension check. To determine conformance to 3.8.3, 3.8.3.1, and 3.8.3.3, the complete suspension system and components shall be checked for proper assembly, installation; whenever seals are used, they shall be checked for proper installation as required to preclude leakage of lubricants and entrance of foreign matter into bearings. Failure to conform to 3.8.3, 3.8.3.1, and 3.8.3.3 shall constitute failure of this test.

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4.5.2.4 Air brake system. Each vehicle, less payload, coupled to prime mover shall be operated over smooth hard-surfaced roads. Vehicle service brakes shall be operated to the extent necessary to assure proper adjustment and performance and conformance to 3.8.5 and 3.8.6. Failure to conform to 3.8.5 and 3.8.6 shall constitute failure of this test.

4.5.2.5 Electrical system test. To determine conformance to 3.8.7 thru 3.8.9, the electrical system and all components shall be tested for proper operation, assembly, mounting, installation and continuity while the vehicle is in operation. Failure to conform to 3.8.7 thru 3.8.9 shall constitute failure of this test.

4.5.2.6 Fording test. Determine conformance to requirements of 3.4.6. Inability to conform to requirements of 3.4.6 shall constitute failure of this test.

4.5.2.7 Stability test. A test shall be performed to demonstrate conformance to 3.4.5. Inability to conform to requirements of 3.4.5 shall constitute failure of this test.

4.5.2.8 Test procedures of lifting provisions.

a. Lift the distributor and hold in suspension in its normal travel position using slings hat converge not more than 24 feet above the lowest extremity of the distributor. Determine the forward angles of application for each provision. Measure the clearance between each sling and the distributor. Measure eye openings and clearance dimensions of the provision. Measure the load carried by each provision.
b. Restrain the distributor by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in a above. An alternate method of test is to lift the distributor as in a above and add weights to the main frame until the required force is obtained. Hold the distributor under either of the above loading conditions for at least 90 seconds. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.9 Test procedures of tiedown provisions. With the distributor anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.10 Vehicle weight. The distributor shall be weighed to determine the curb weight and distribution of the curb weight on the kingpin and rear axles. The distributor with the tank filled with water shall be weighed to determine the load on the kingpin and rear axles. Load in excess of 30,000 pounds on the kingpin or load in excess of any rear suspension component shall constitute failure of this test.

4.5.2.11 Pumping system operations. Perform the following functions:

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a. Start engine from remote control panel. Shut engine down from remote control panel.
b. Start engine from remote control panel. Shut engine down from engine control panel.
c. Start engine from engine control panel. Shut engine down from engine control panel.
d. Start engine from engine control panel. Shut engine down from remote control panel.
e. When performing a thru d determine if pilot lights are functioning properly.
f. Pump water (through the 30-foot filling hose) from an external source into the tank using the distributor pump.
g. Pump from the tank through the spray nozzles. During this phase demonstrate the independent control of each nozzle valve. Determine the width of coverage (both nozzles on and single nozzle on), spray pattern and spray overlap.
h. Pump water from an external source through the fire hose.
i. Pump water from the tank through the fire hose.
j. Drain tank without using pump.

Inability to perform any of these operations in accordance with requirements specified herein shall constitute failure of this test.

4.5.2.12 Pump performance test. Perform tests in accordance with the applicable provisions of the test code for centrifugal pumps of the Standards of the Hydraulic Institute. Correct all data to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68° F. Inability of the pump to meet the requirements of 3.7.1 shall constitute failure of this test. For test purposes all data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot at a temperature of 68° F.) A failure is defined as any malfunction which cannot be corrected within 60 minutes (all waiting periods excluded) by adjustment using controls and on-equipment tools available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation or degradation of performance below specified levels.
b. Damage to the pump assembly by continued operation.
c. Safety hazards to personnel.

4.5.2.13 Operation. Operate the pump continuously at a total head capacity commensurate with the continuous horsepower rating of the engine for a period of 100 hours. During this period, only engine adjustments and normal scheduled maintenance shall be permitted. Examine the pump at the end of each 8-hour period. A maximum shut-down time of 30 minutes during the 8 hours is allowable for maintenance and examination. During operation, determinations shall be made as to whether there is leakage through the pump case or at the pump shaft seal

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and whether there is malfunction of any component. Evidence of leakage at any time, malfunction of any component, or inability of the pump to operate continuously as specified above shall constitute failure of this test.

4.5.2.14 Tracking tolerance, jackknife and landing leg test. During the mileage and speed test (table II) with the distributor filled perform a through c below:

a. Measure and record the deviation or off tracking of the distributor with respect to the prime mover, when the distributor is towed straight ahead on a paved surface. Failure to conform to allowable DOT tracking tolerances shall constitute failure of this test.
b. Jackknife the distributor at angles up to 90 degrees from straight ahead both to the left and to the right. Inability to jackknife at least 90 degrees in either direction shall constitute failure of this test.
c. At the end of the road test connect and disconnect the distributor from the truck tractor at least two times. After each disconnect, raise and lower the fifth wheel height by use of the adjustable landing legs. Inspect the landing legs for permanent deformation or binding which constitutes failure. Inspect tank, suspension, brakes, plumbing and spraying system, accessories, and hoses. Permanent deformation, weld cracks, air leaks, missing or loose accessories shall constitute failure of this test.

4.5.2.15 Noise level test. The equipment noise level at the operator's station and all immediate occupied areas shall not exceed 85 dB(A) when tested in accordance with SAE J919. Exterior noise levels for drive-by shall not exceed 88 dB(A) when tested in accordance with SAE J 88. If the data exceeds the limits specified in category D, table 2, of MIL-STD-1474, for equipment noise level, or 88 dB(A) for drive-by, the contractor shall reduce the noise level or provide documentation verifying that compliance is either technically or economically unfeasible.

4.5.2.16 Spray distributor rate test. The distributor shall be tested to determine the accuracy of the distribution chart specified in 3.7.4h. Inability to demonstrate an accuracy of ±15 percent, for each variable, shall constitute failure of this test.

4.5.2.17 Spraying mode test. The distributor shall be tested as follows:

a. Connect the distributor to the truck tractor.
b. Remove the remote control from the distributor and mount on truck tractor.
c. Fill the distributor to capacity from a pond, river, or lake using the two 15-foot sections of suction hose.
d. Proceed to spray area. Spray water with distributor in motion through both nozzles at maximum flow rate, until low level warning light comes on.
e. Return to loading point.
f. Remove the remote control from the truck tractor and mount on distributor.

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g. Repeat steps b, c, d, e, and f until 10 tank loads have been sprayed.

Inability to perform any of these operations or evidence of any failure or malfunction shall constitute failure of this test.

4.5.2.18 Tractor distributor compatibility check. Determine conformance to requirements of 3.4.2a. Inability to conform to requirements of 3.4.2a shall constitute failure of this check.

4.6 Toxicological data and formulations inspection. The contractor shall have available to the contracting officer a certificate of compliance that the liners and all surfaces that contact the water are non-toxic (see 3.6). The certificate shall be supplied by an independent material testing laboratory or by the product manufacturer.

4.7 Inspection comparison. The Government may select water distributors at any time during the contract production period and subject them to any test or examination as specified in section 4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units-selected at random from those which have been accepted by the Government and will not include the previously inspected first article vehicle.

4.7.1 Inspection failure. Failure of an inspection comparison water distributor to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.5, shall be cause for rejection of the inspection comparison water distributor(s), and shall be cause for refusal by the Government to continue acceptance of production water distributors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies.

4.8 Inspection of packaging. Prior to shipment, all vehicles produced shall be examined for preservation and marking which shall be in accordance with the quality assurance provisions of MIL-D-771.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete distributor shall be preserved, packed and marked in accordance with MIL-D-771. The degree of preservation shall be A, B, or commercial and the degree of packing shall be A, B, or commercial as specified (see 6.2). When the commercial packaging option of MIL-D-771 is selected, preservation and packing shall comply with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The distributor is intended for use in soil stabilization and soil compaction operations requiring distribution of water. The distributor can also be used for dust alleviation, fire fighting and transport water for emergency use.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When a first article is required (see 3.2).
d. Time frame required for submission of first article model.
e. When the Government will conduct any or all of the first article model examinations and tests. When the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).
f. Type and number of technical publications to be furnished (see 3.17).
g. The repair parts and maintenance tools to be furnished (see 3.18).
h. When certification is required (see 3.22).
i. Degree of preservation and packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first articles.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.19.

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